

Student Contribution Report

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Project: NBP Analyzer

Team: BunniHopPon

Role: Developer

Overview

I am a member of the BunniHopPon project group, responsible for the development of the NBP Analyzer system. My official role in the team is Developer. My primary focus was on implementing core mathematical and rate-alignment logic, visual design layout polishing, co-developing the user interface for session trends, and setting up testing environments.

Key contributions

1. Core Math Logic & Currency Analytics

Trend Metrics Calculations: Developed statistical calculations in *sessionTrends.js*:

- *calcMinMaxAvg*: Computes the minimum, maximum, and average exchange rates.
- *calcCoeffOfVariation*: Computes the coefficient of variation using standard deviation and mean.

Numerical Precision Fixes: Resolved floating-point bugs in exchange rates change percentage and trend math formulas.

2. Network & API Enhancements

- API Request Chunking: Implemented *fetchNbpRates* and *chunkDateRange* to split large user date-range requests (e.g. 1 year) into chunks of maximum 93 days, sending queries in parallel and merging them seamlessly.
- Implemented checks in data loading catches to display custom friendly error banners instead of uncaught exception logs.

3. UI/UX Development & Polishing

- Co-Implementation of Session Trends Panel: Collaborated on building and visual formatting of the *SessionTrendsPanel* module, focusing on:

- **Component Architecture:** Co-developed and structured the panel layout to provide a clean interface for currency and period selection.
- **Data Visualization:** Integrated the Reactcharts/ChartJS library to visualize exchange rate fluctuations via line charts.
- **Selector Implementation:** Developed custom currency and date selectors to capture user input parameters.
- **Loading States:** Implemented loading spinners and interactive control disabling during data fetches to improve user experience.
- **Logo Branding:** Mounted the project logo image beside the main header in *App.jsx*.
- **Control Labels & Spacing:**
 - Added the "Time period" section labels to selectors and the "Rising, falling, and unchanged sessions analysis" diagram description header in the *SessionTrendsPanel*.
 - Standardized the "Base currency", "Quote currency", and "Calculation period" labels on the *HistogramPanel* for visual continuity.
 - Unified the bottom footer padding to resolve double-padding bugs.

4. Unit Tests

Developed and executed unit test suites in *changeDistribution.test.js* to verify core calculations:

- **calculateDailyChanges tests:** Verifies error handling for invalid arrays, small datasets (<2 points), and negative/invalid rates, alongside float-precision change calculation checks.
- **initializeBins tests:** Verifies map creation with the 13 default frequency range bins between -3.0% and +3.0%.
- **determineBin tests:** Validates correct boundary keys, exact limit adjustments, and safe zero value handling.
- **groupChangesIntoBins tests:** Asserts proper bin frequency incrementing and dynamic out-of-bounds bin creation.
- **getHistogramDistribution tests:** Runs end-to-end integration tests using mocked NBP API responses.